

Interactive Web: Leveraging AI-Driven Code Generation to Create 3D Venous Valve for DVT Risk Analysis

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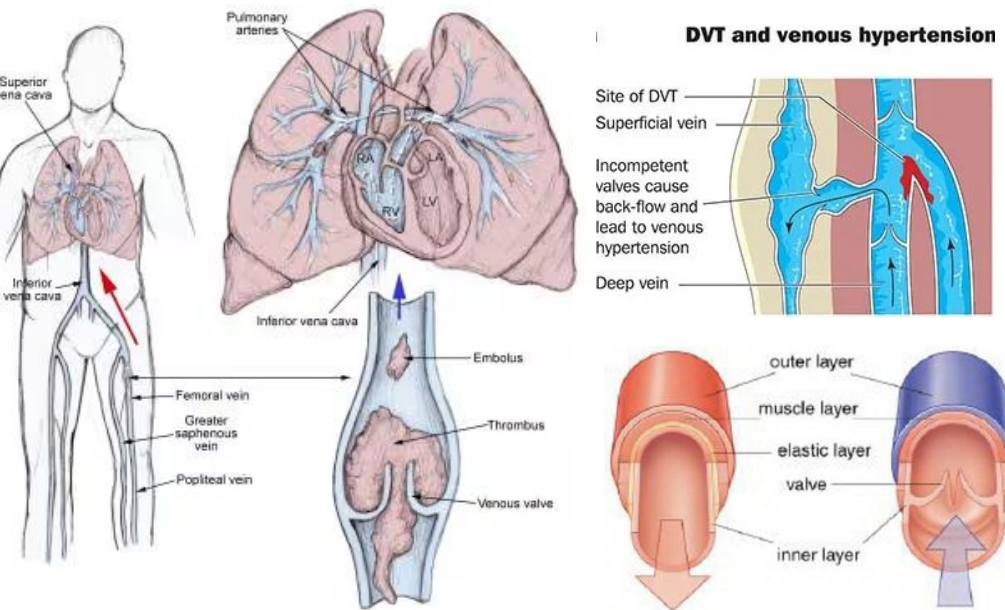
MOREAU
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COMPASSIONATE COLLEGE PREP



INTRODUCTION

Deep vein thrombosis (DVT) affects an estimated 900,000 Americans annually, with venous valve dysfunction playing a central role in thrombogenesis. Current clinical assessment relies on operator-dependent ultrasound interpretation, introducing significant variability and limiting scalability. This study presents an agentic AI framework that autonomously analyzes venous valve dynamics from ultrasound sequences, moving beyond static geometric measurements to capture time-resolved mechanical behaviors linked to DVT risk. The system classifies valve competence and flags hemodynamic risk states without manual intervention, offering a reproducible, scalable alternative to conventional assessment.

BACKGROUND

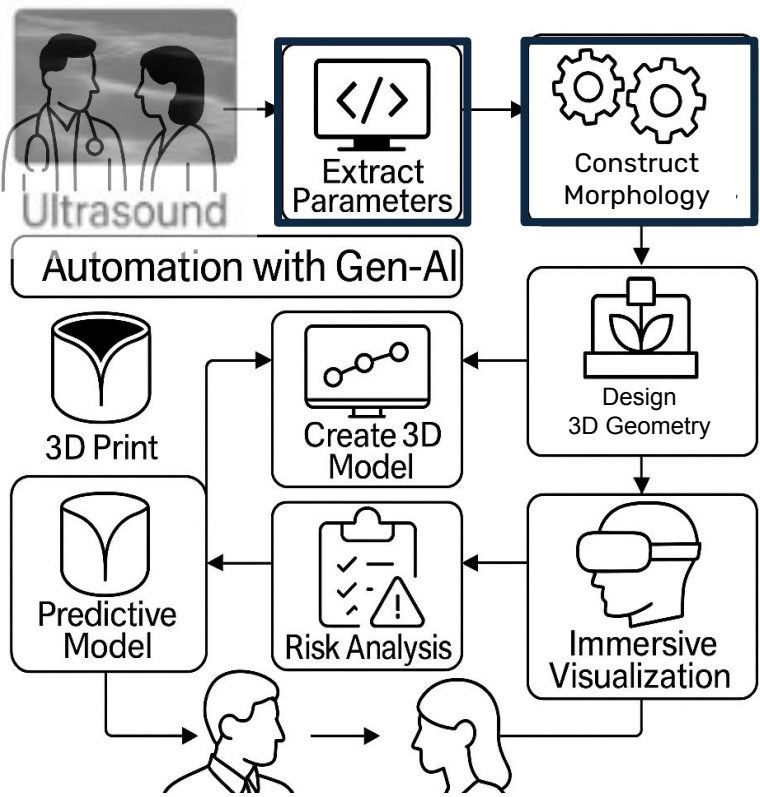


Goal: Autonomously classify venous valve mechanical states (normal vs. abnormal) from ultrasound sequences by extracting time-resolved geometric features — leaflet opening angles, asymmetry indices, and cycle-to-cycle variability.

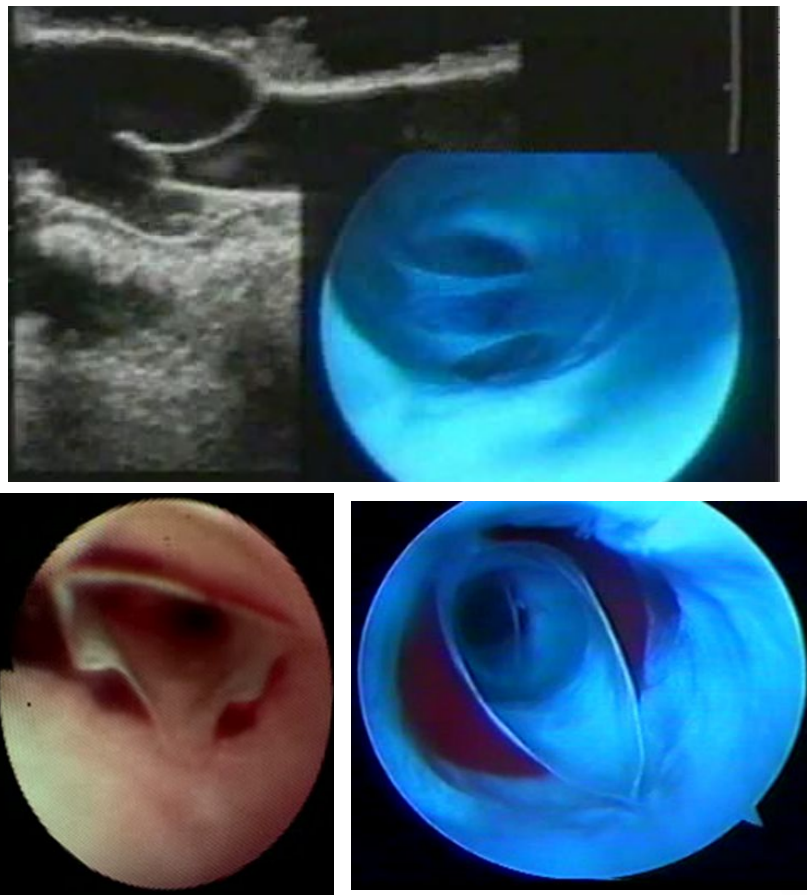
Hypothesis: An agentic AI framework can detect pathological valve behaviors (partial openings, irregular closing rates) that generate flow separation and low-shear recirculation zones predictive of DVT risk, with accuracy comparable to expert annotation.

Purpose: Replace operator-dependent manual assessment with a scalable, data-driven pipeline for valve competence classification and personalized DVT risk stratification.

METHODOLOGY



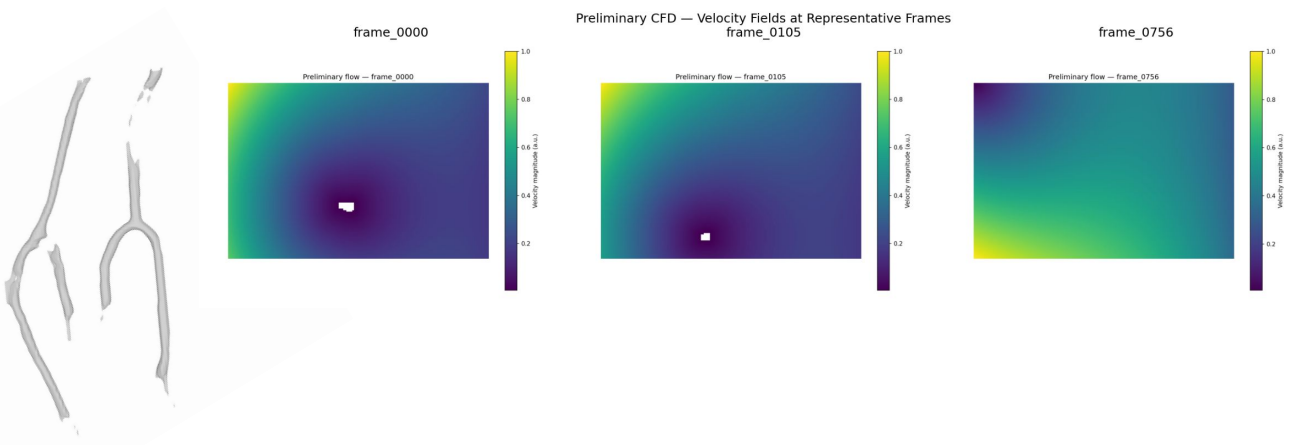
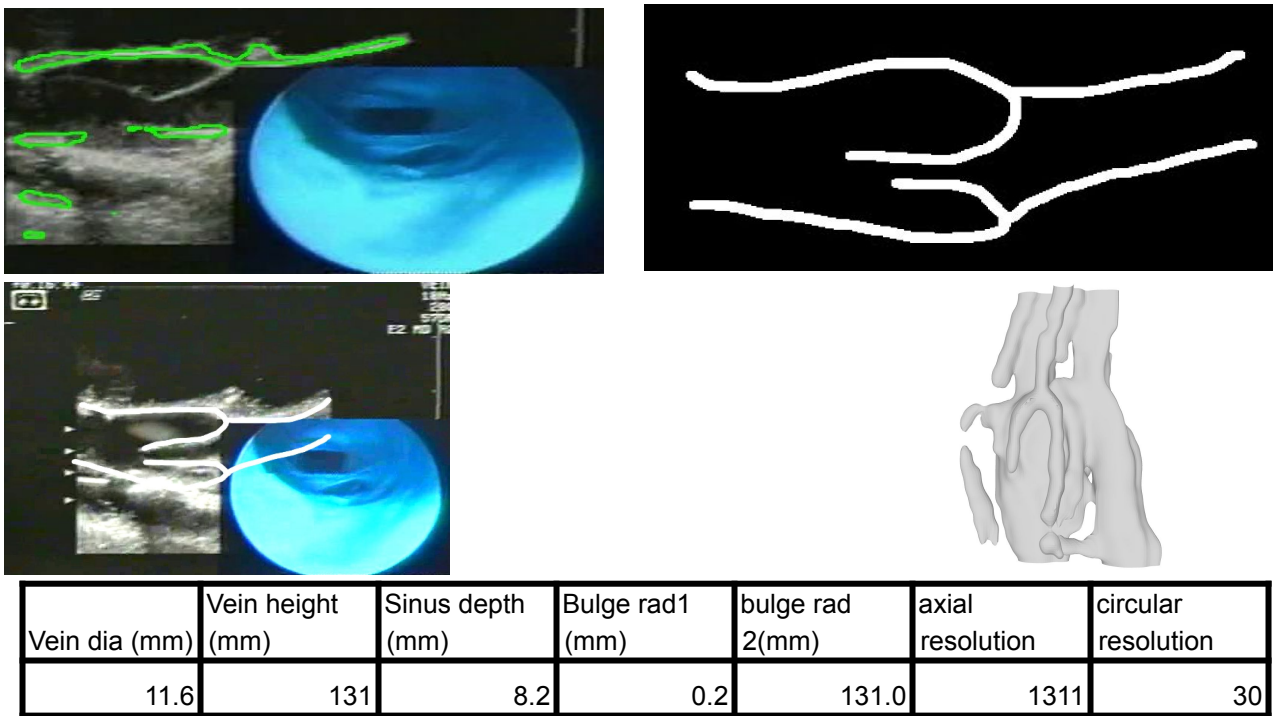
PROBLEM STATEMENT



Top: Ultrasound image of a venous valve alongside a 3D-rendered view showing internal leaflet geometry. **Bottom left:** Endoscopic or simulated view of a closed valve. **Bottom right:** Valve in a partially open state, highlighting leaflet curvature and flow direction.

PROCESS

RESULTS



DISCUSSION

This work introduces an autonomous, agentic AI framework that addresses the core limitation of current DVT risk assessment: its dependence on expert, operator-driven interpretation. By targeting time-resolved valve dynamics rather than static anatomy, the system captures the mechanical signatures most directly linked to thrombogenesis, specifically partial leaflet openings and irregular closing rates that produce the low-shear recirculation environments where clot formation initiates. The pipeline generalizes from limited annotated data through transfer learning, reducing the burden of expert labeling at scale. Unlike prior approaches focused on geometric reconstruction, this framework outputs clinically interpretable risk classifications directly from raw ultrasound input. Integration with patient-specific CFD models (via DeepVein Inc.) is a natural next step, enabling the classified valve states to drive downstream hemodynamic simulation and risk quantification.

CONCLUSION

- We successfully built a tool that converts ultrasound videos of veins into 3D STL models. After extracting frames, users manually annotate a few of the frames, which are used to train a U-NET model for edge detection. The detected edges are then processed into a clean 3D mesh of the venous valve.
- A key challenge was accurately modeling valve structures from noisy ultrasound data. Our pipeline was able to generate the 3D model from the outside but was unable to capture the valve leaflets on the inside (Figure).
- In the future, we plan to add VR and better tools for immersion as well as for better analyzing risk analysis for DVT.

REFERENCES

1. HUMAN BIOLOGY / BLOOD VESSELS - Pathwayz. (n.d.). <https://www.pathwayz.org/Tree/Plain/BLOOD+VESSELS>
2. Ouellette, D. R. (2019, April 29). Pulmonary Embolism: Practice Essentials, Background, Anatomy. Medscape.com. <https://emedicine.medscape.com/article/300901-overview>
3. Seckler, D. J. (2017). Understanding Deep Vein Thrombosis (DVT) - King Air. King Air. <https://kingairmagazine.com/article/understanding-deep-vein-thrombosis-dvt>
4. Buescher, C. D., Nachiappan, B., Brumbaugh, J. M., Hoo, K. A., & Janssen, H. F. (2005). Experimental studies of the effects of abnormal venous valves on fluid flow. Biotechnology progress, 21(3), 938–945. <https://doi.org/10.1021/bp049835u>
5. Zhang, K., Kumar, V., & Zhang, M. (2025). Interactive Web: Leveraging AIDriven Code Generation to Simplify Web Development Algorithms for Novice Programmers (pp. 81–91). <https://doi.org/10.5121/csit.2024.150107>

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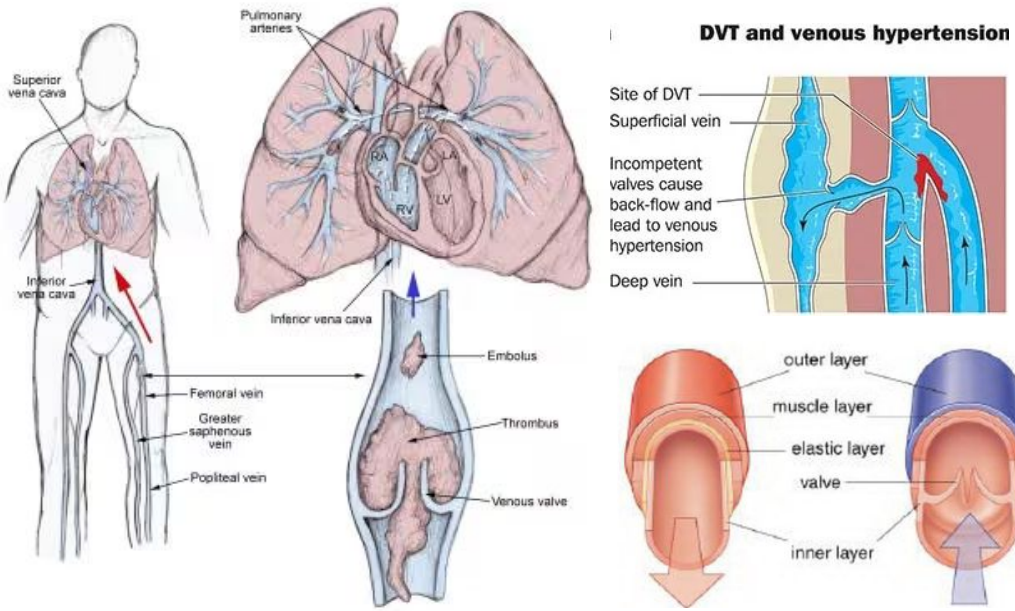
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INTRODUCTION

This study presents a generative **AI-based framework** for creating **3D models of venous valves** to investigate deep vein thrombosis (DVT) risk factors. Using a Ultrasound videos as an input, the framework is designed to create *in-silico* models (STL files) of the valve. These models can be used for: 3D printed replica, predictive analysis (e.g. computational fluid dynamics or CFD analysis), creating a prosthesis valve, or better diagnosis and treatment plans. By automating geometry generation, the framework significantly enhances scalability and prototyping efficiency. While the current focus is on anatomically accurate venous valves, the approach offers broad potential for personalized biomedical applications and streamlined clinical research workflows.

BACKGROUND

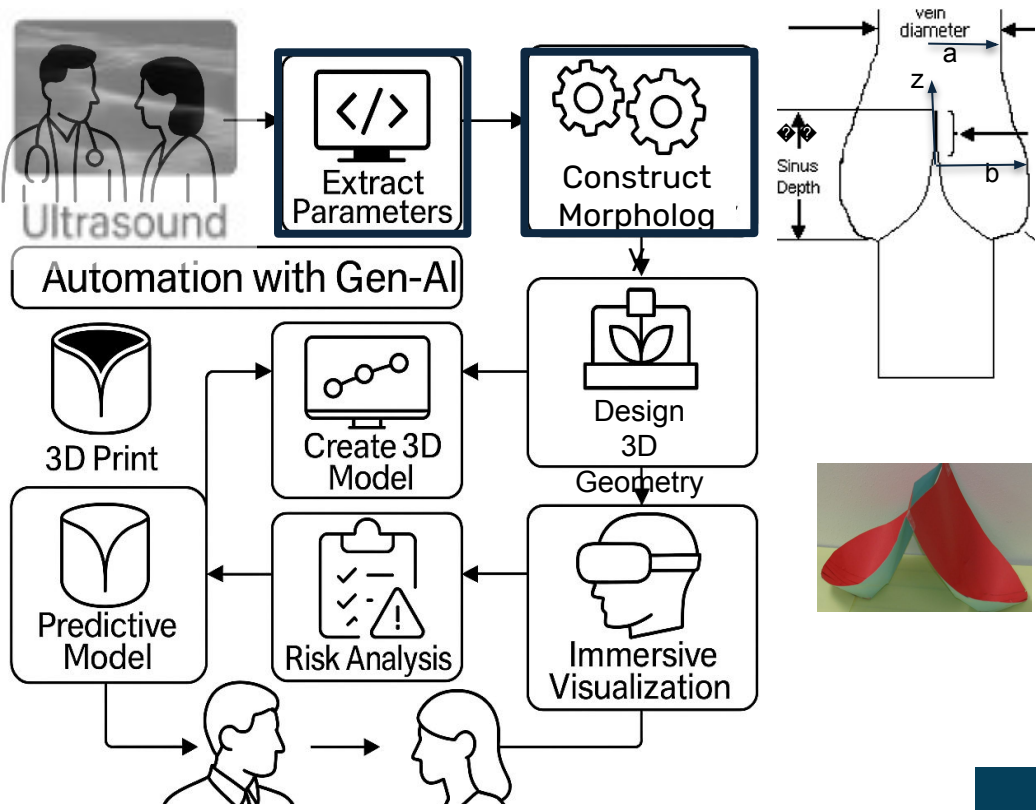


Goal: Extract important parameters (Vein diameter, Sinus depth, Bulging Vein Wall Diameter, Leaflet material properties, etc.).

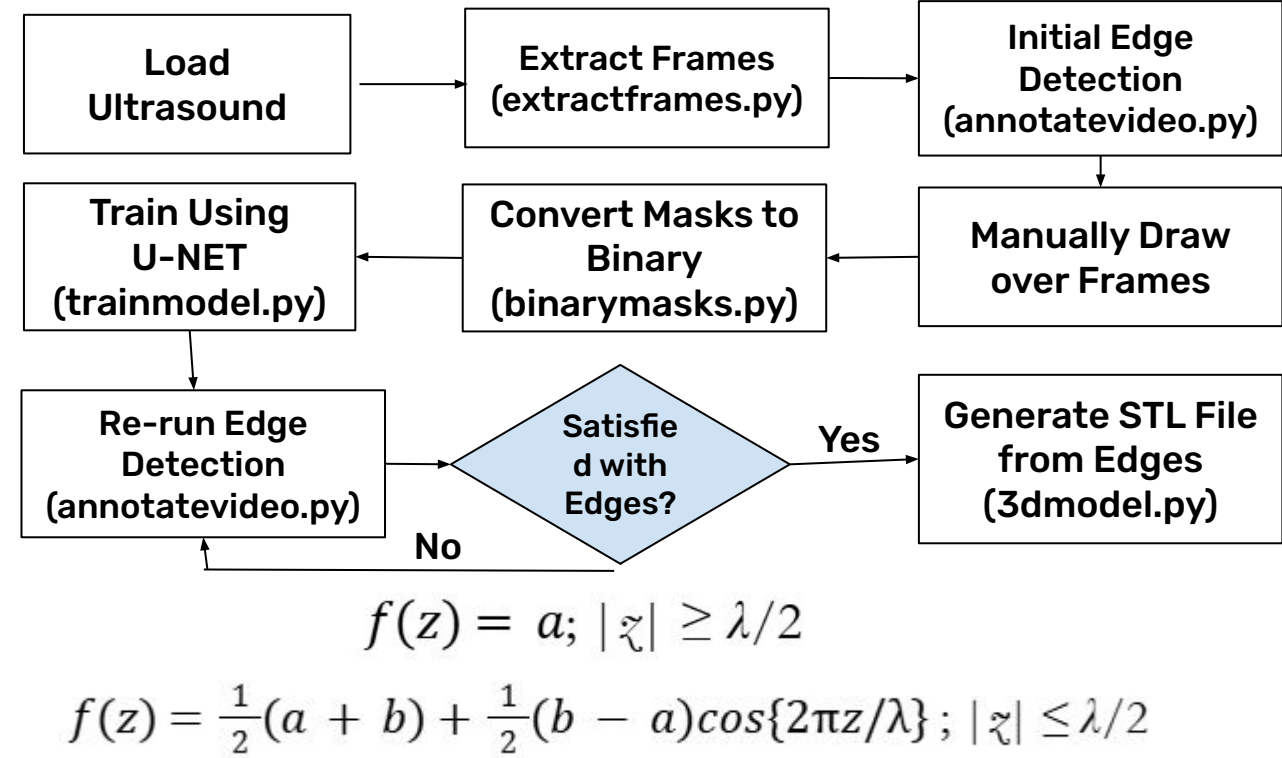
Hypothesis: Leading edge ML (e.g., GenAI/ChatGPT) can be used to automatically extract valve morphological characteristics to an accurate 3D of a patient's venous valve from Ultrasound image in real-time.

Purpose: Perform predictive analysis to quantify DVT risks for personalized diagnosis and preventive care.

METHODOLOGY

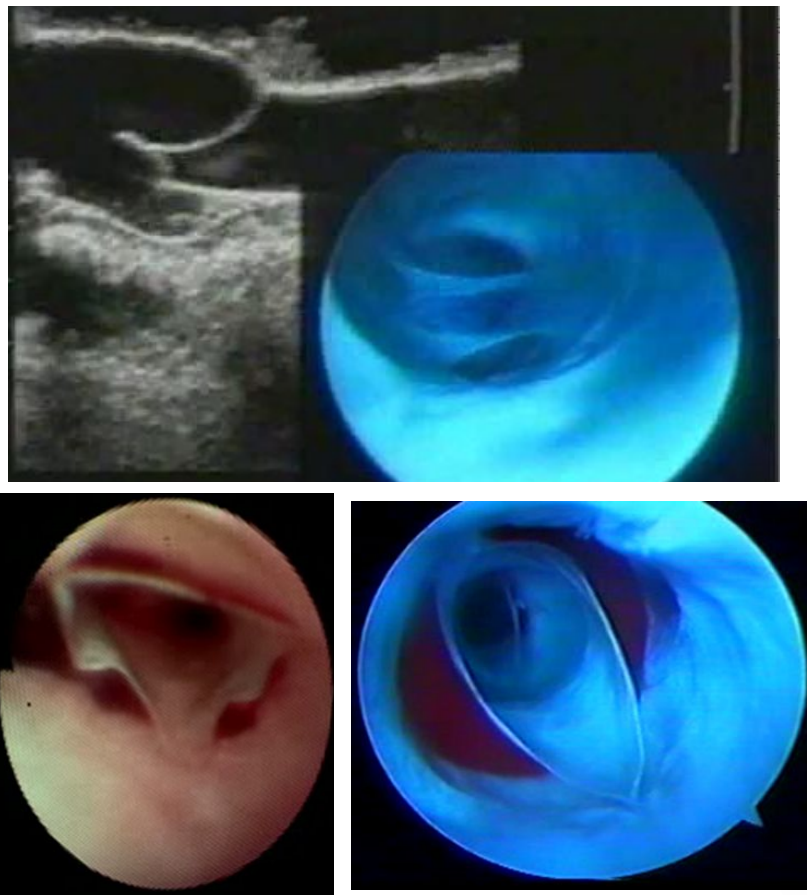


PROCESS



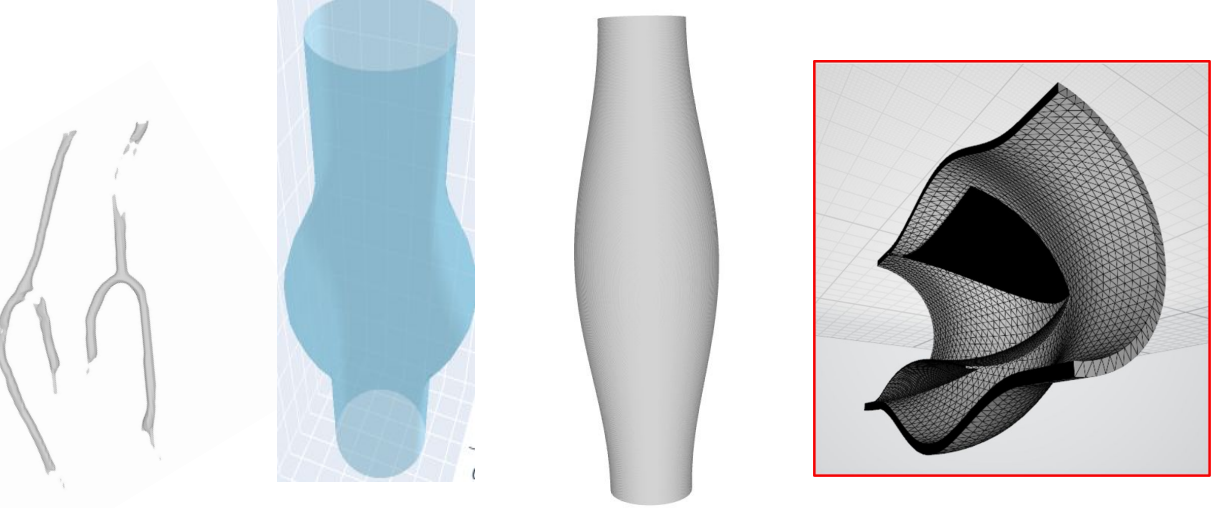
RESULTS

PROBLEM STATEMENT



Top: Ultrasound image of a venous valve alongside a 3D-rendered view showing internal leaflet geometry. **Bottom left:** Endoscopic or simulated view of a closed valve. **Bottom right:** Valve in a partially open state, highlighting leaflet curvature and flow direction.

Vein dia (mm)	Vein height (mm)	Sinus depth (mm)	Bulge rad1 (mm)	bulge rad 2(mm)	axial resolution	circular resolution
11.6	131	8.2	0.2	131.0	1311	30



DISCUSSION

This project introduces a machine learning–driven framework for automating the creation of 3D venous valve models from ultrasound videos. Using PyTorch and a custom-trained U-NET architecture, the system detects valve boundaries from annotated ultrasound frames. These detected edges are processed into binary masks, which are then converted into STL files via a Python-based geometry engine. The approach minimizes manual intervention by leveraging deep learning to generalize from limited annotations, enabling scalable, anatomically relevant model generation. While current leaflet morphology is simplified, the pipeline provides a robust foundation for integrating CFD analysis and patient-specific modeling.

CONCLUSION

- We successfully built a tool that converts ultrasound videos of veins into 3D STL models. After extracting frames, users manually annotate a few of the frames, which are used to train a U-NET model for edge detection. The detected edges are then processed into a clean 3D mesh of the venous valve.
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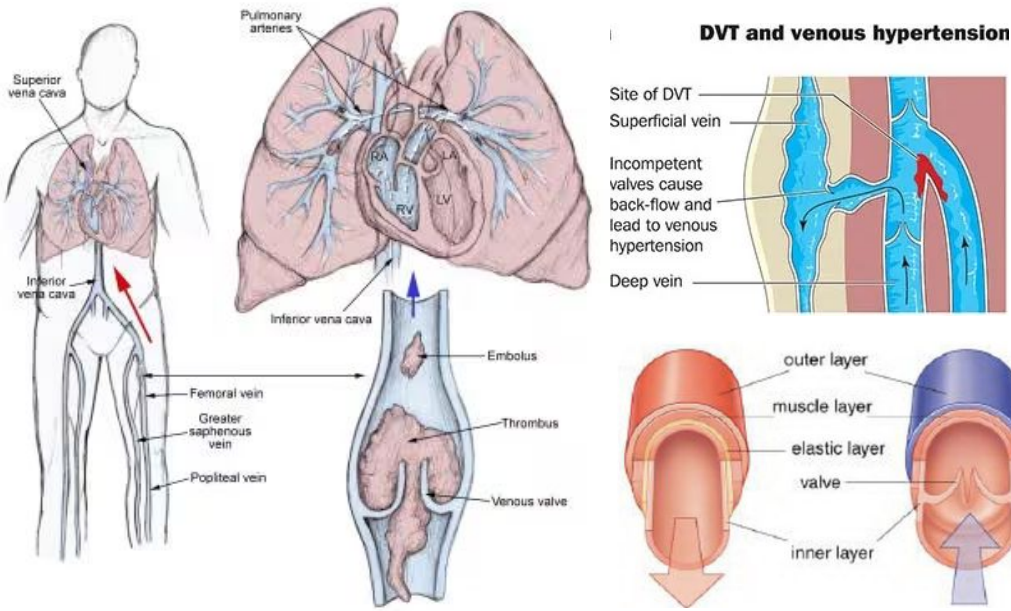
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BACKGROUND

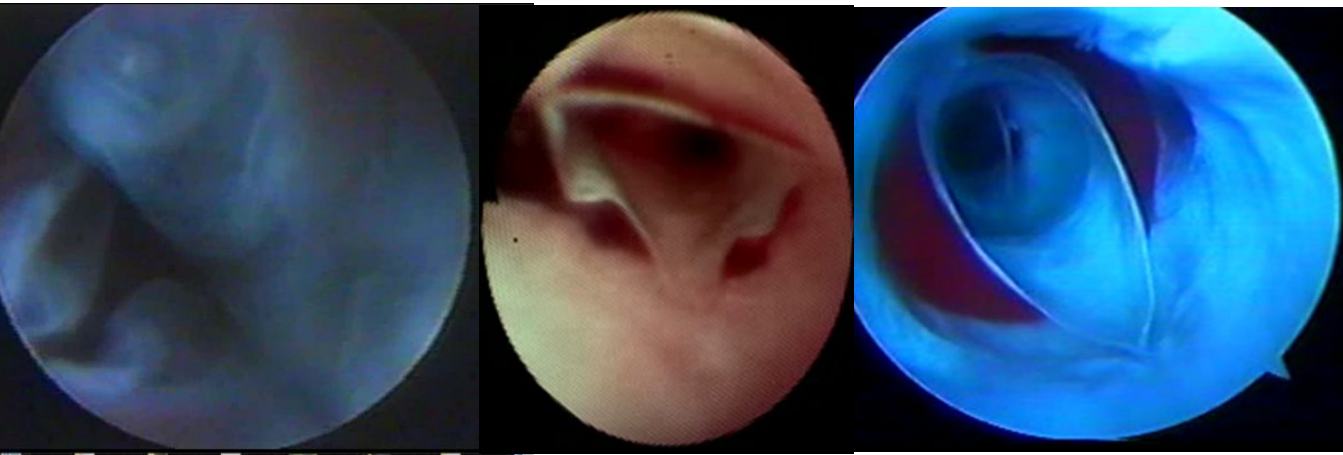
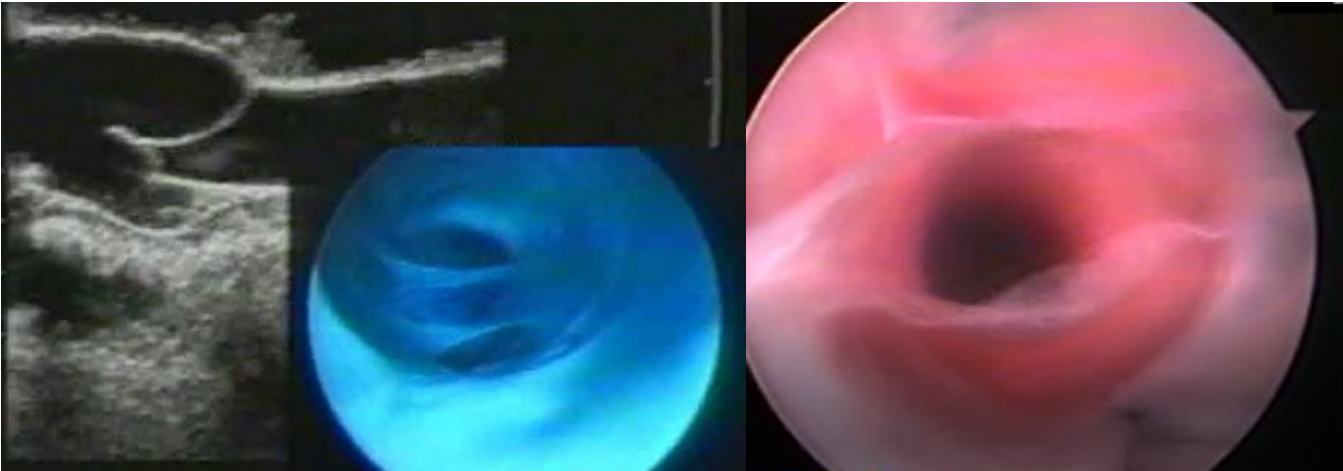


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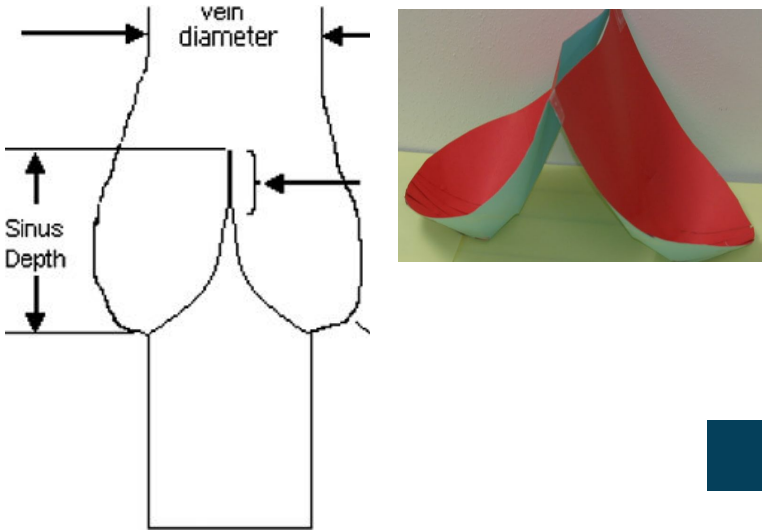
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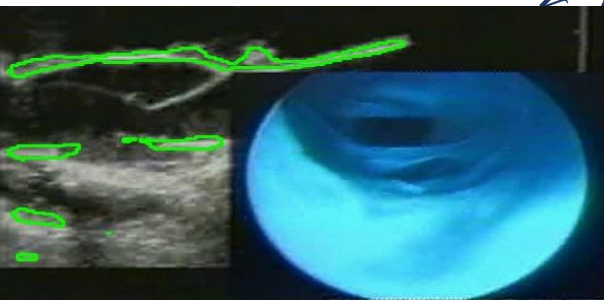
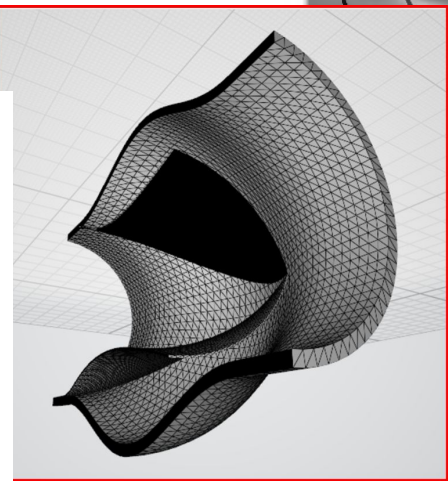
METHODS



RESULTS



$$f(x) = \frac{1}{2}(a + b) + \frac{1}{2}(b - a)\cos(2\pi x/\lambda); |x| \leq \lambda/2$$



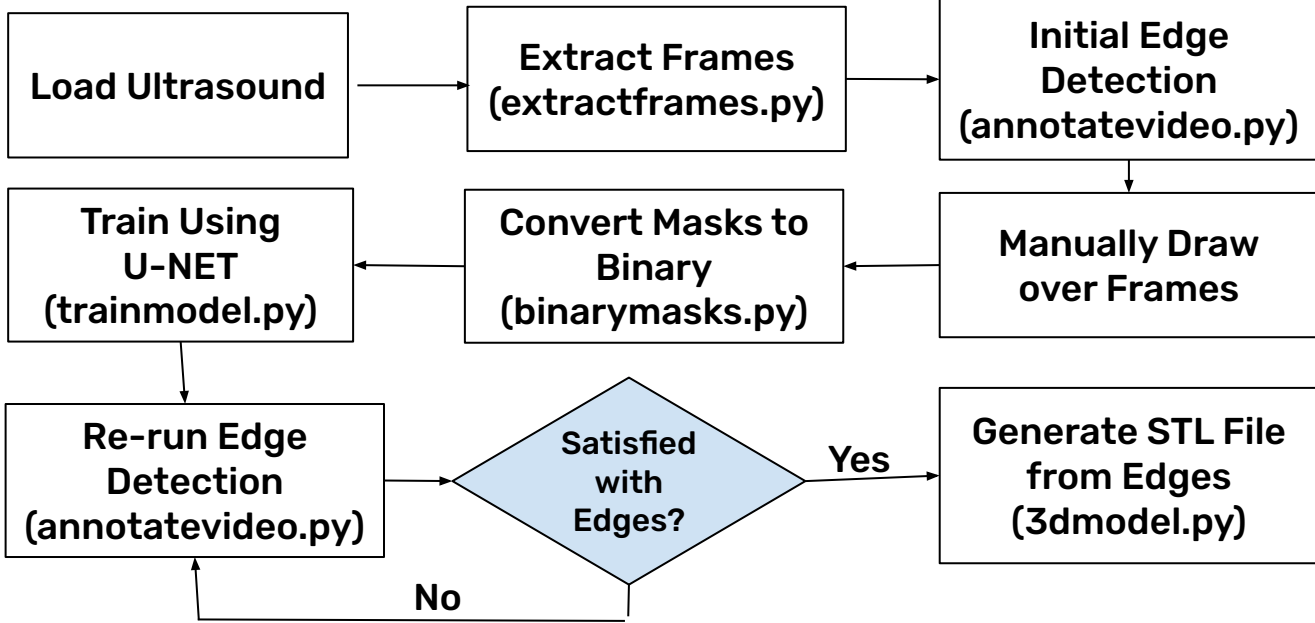
This project presents a generative AI-driven framework for automating the design of 3D venous valve models. By combining GPT-4-based parameter generation with a Python-based geometry engine, the system

PROBLEM DEFINITION

printing and simulation. The approach significantly reduces manual design time, enabling rapid prototyping with anatomically realistic variability. All generated models were validated as structurally sound and compatible with common biomedical analysis tools. While current leaflet designs are simplified, the workflow establishes a scalable

with

PROCESS



REFERENCES

1. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111111/>
2. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111111/>
3. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111111/>
4. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6111111/>

ePoster Template

Template contains:

- 5-column page w/placeholders; 5-column page w/headers only
- 4-column page w/placeholders; 4-column page w/headers only
- 3-column page w/placeholders; 3-column page w/headers only

Select “New Slide”

Choose which format you’d like to use

To make edits to the format, go to the View tab and choose “Slide Master”

Poster Template / Print Size:

This PowerPoint template will be used for both digital and printed posters at the ASAIO Annual Meeting. The slide size is set at 24in x 13.5in which is in the 16:9 ratio and will fill the screen on the ePoster monitors.

Poster should be printed at 200% with a final size of 48in wide x 27in high

The display stands for your printed posters are 8ft x 4ft with 2 posters per board.

Image Quality:

You can place digital photos or logo art in your poster file by selecting the **Insert, Picture** command, or by using standard copy & paste. For best results, all graphic elements should be at least **150-200 pixels per inch in their final printed size**. For instance, a 1600 x 1200 pixel photo will usually look fine up to 8in x 10in wide on your printed poster.

To preview the print quality of images, select a magnification of 200% when previewing your poster. This will give you a good idea of what it will look like in print.

Please note that graphics from websites (*such as the logo on your hospital or university home page*) may only be 72dpi and not suitable for printing.

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SAMPLE



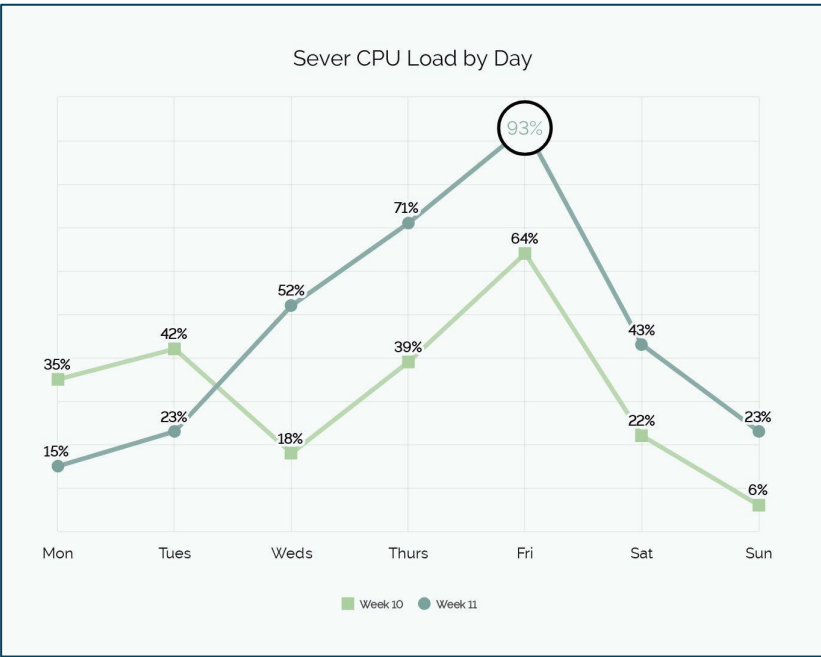
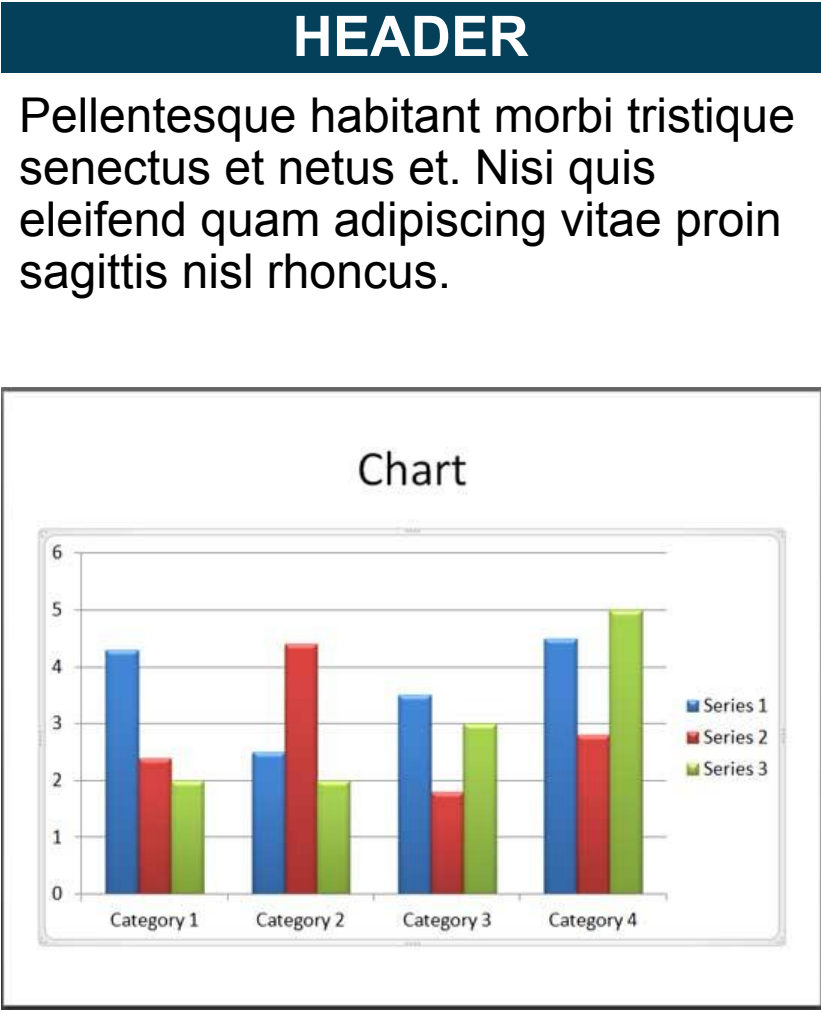
HEADER

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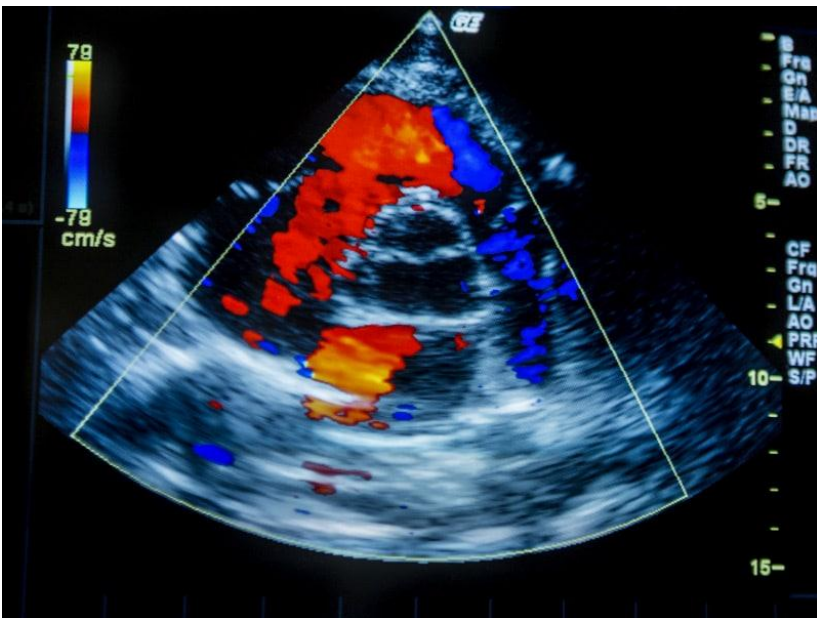
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HEADER

Cookie Sales by Region			
SalesRep	Region	# Orders	Total Sales
Bill	West	217	\$41,107
Frank	West	268	\$72,707
Harry	North	224	\$41,676
Janet	North	286	\$87,858
Joe	South	226	\$45,606
Martha	East	228	\$49,017
Mary	West	234	\$57,967
Ralph	East	267	\$70,702
Sam	East	279	\$77,738
Tom	South	261	\$69,496

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Sample 4 Column Template – Title of Presentation/ePoster

Author One, Author Two, Author Three, Author Four, Author 5
Name of Hospital/Institution

SAMPLE



INTRODUCTION

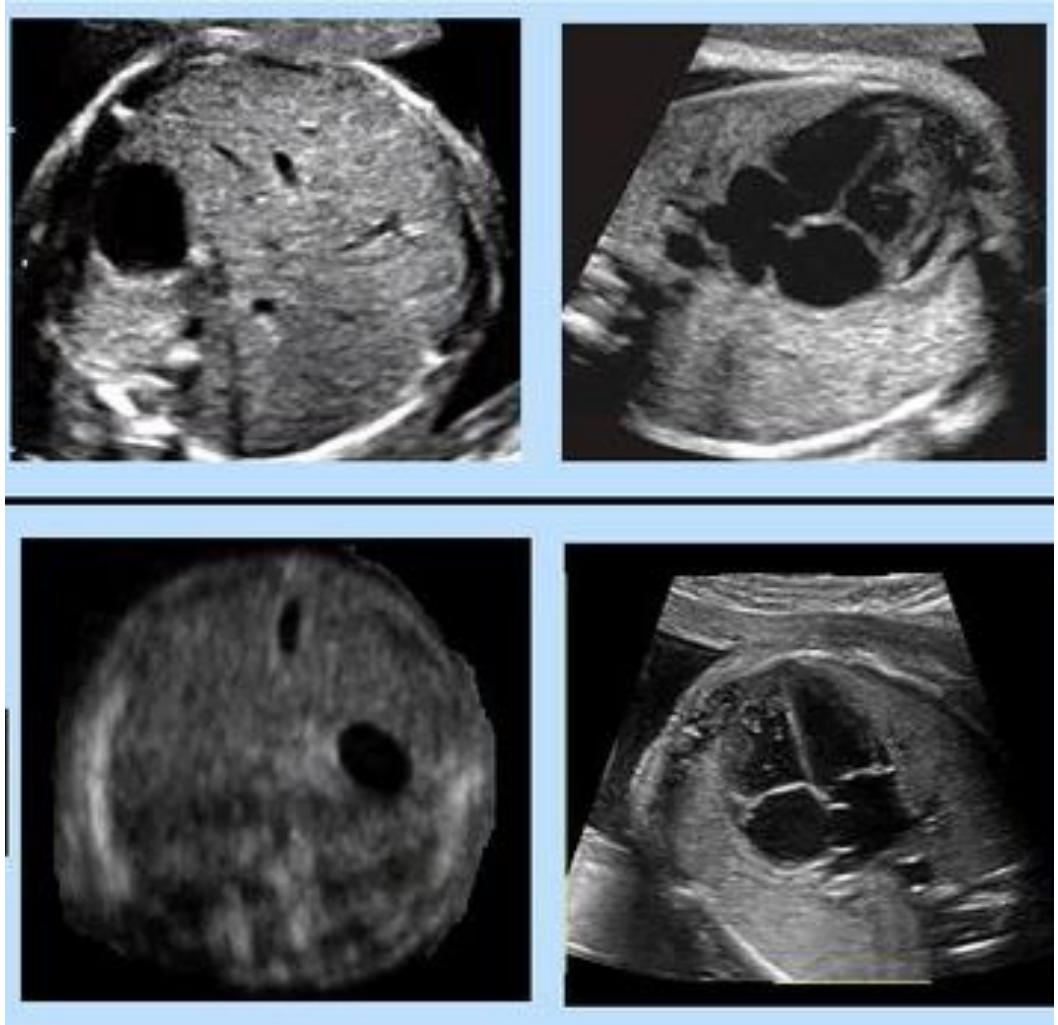
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METHODS

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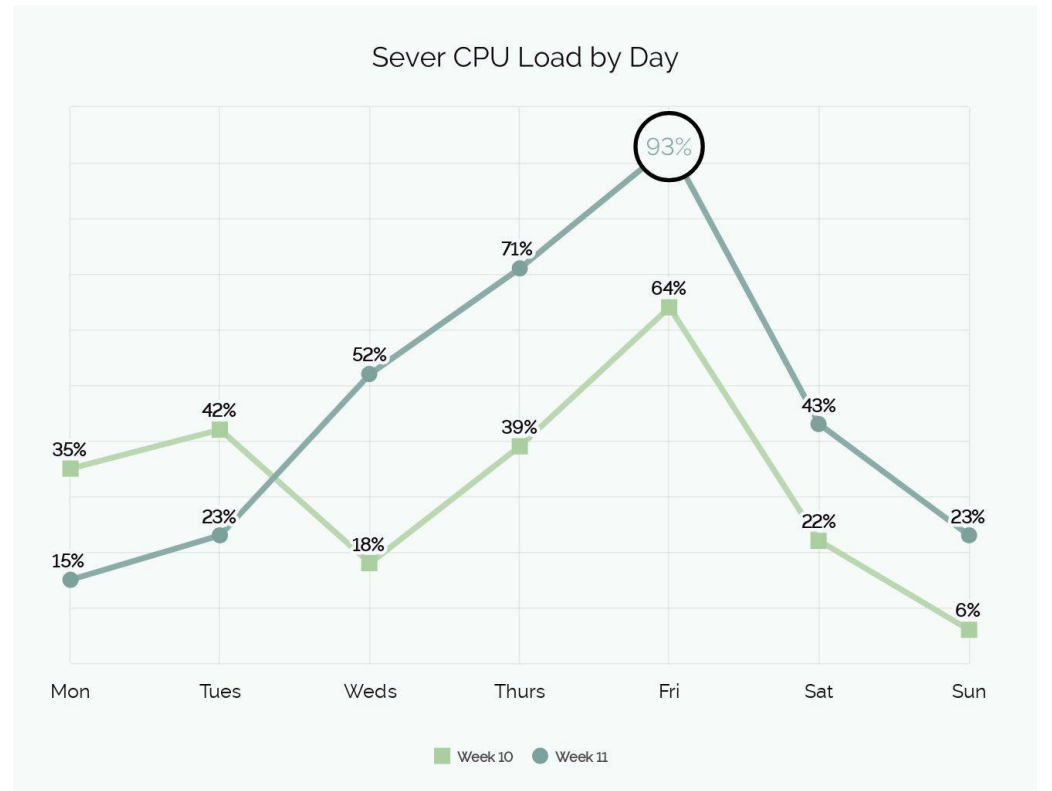
RESULTS

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RESULTS

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DISCUSSION

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CONCLUSION

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Sample 3 Column Template – Title of Presentation/ePoster

Author One, Author Two, Author Three, Author Four, Author 5
Name of Hospital/Institution

SAMPLE

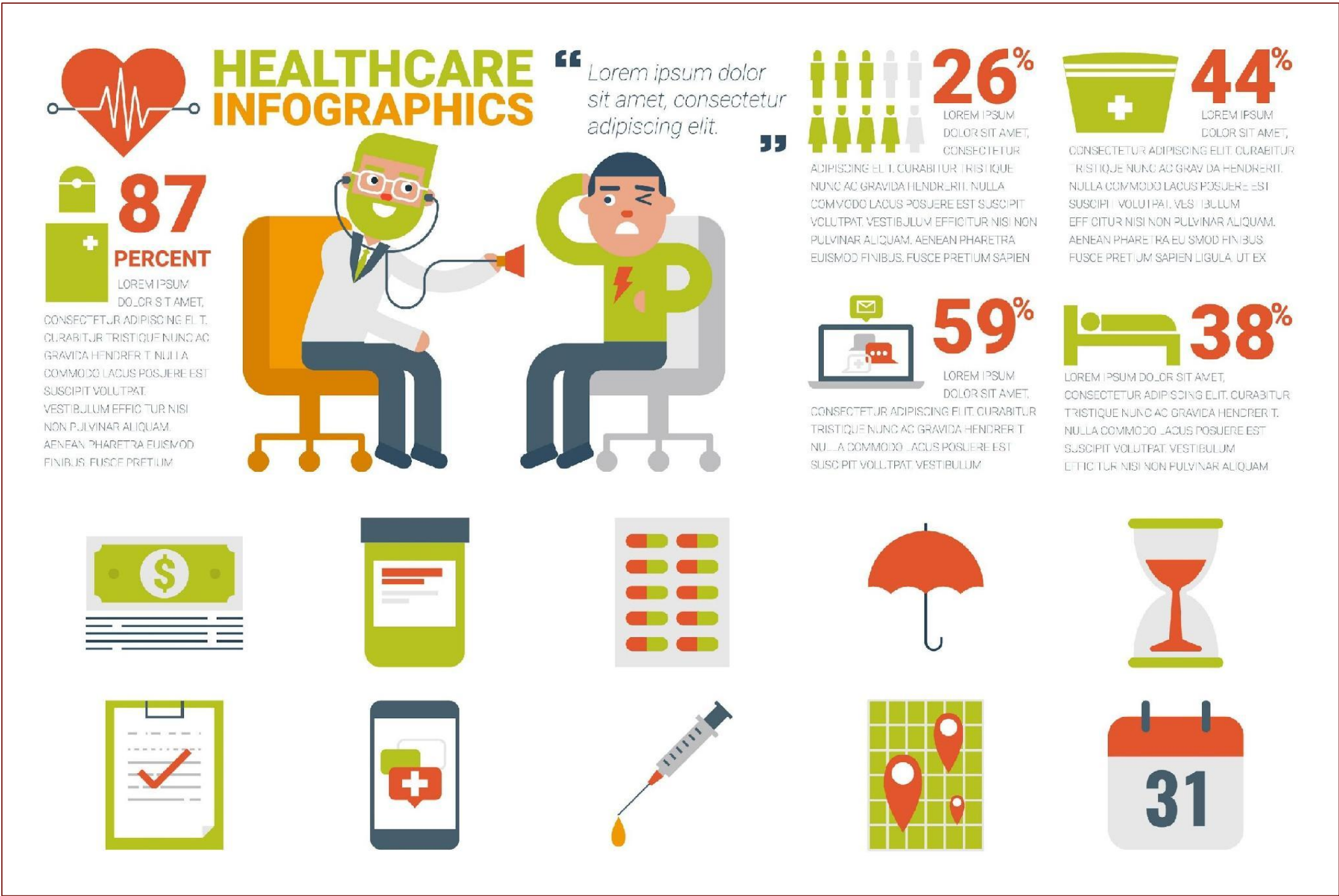


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Abstract

The GYRO implantable Bi-Ventricular Assist Device (BVAD), under development at Baylor College of Medicine, TX, has been studied. The NIH (Normalized Index of Hemolysis) in the pump is measured at three physiological operating conditions. The three-dimensional, time-dependent flow in the pump is modeled by the Navier-Stokes equations, and solved by the finite element method. Hemolysis is predicted along the pathlines using two hemolysis models: the traditional stress-based model and a recent model based on tracking the strain (tensor) experienced by the Red Blood Cells (RBCs) (strain-based model). The strain-based model accounts for physical properties of the RBCs, including their relaxation time (~200 ms) and flow-induced tank-treading motion of their membrane. The hemolysis predicted by the strain-based model is within the error margins of the experimental values, while the stress-based model predicts substantially higher hemolysis. Moreover, some key features of the pump design are examined by various pathlines. It is found that at all flowrates the average residence time along the calculated pathlines is lower than the expected value based on flowrate and pump volume. This is explained by noting that some portions of the pump are hardly accessible to the bulk flow.

Experiments

Mock loop hemolysis experiments are performed as per the ASTM standards (ASTM-F1997). Figure 1 shows the mock loop circuit.

Hemolysis was measured in three typical operating conditions of the GYRO implantable blood pump. Table 1 lists the three operating conditions--A, B and C.

All experiments were done in the horizontal top-contact configuration of the pump. Figure 2 shows the horizontal top-contact configuration of the GYRO pump.

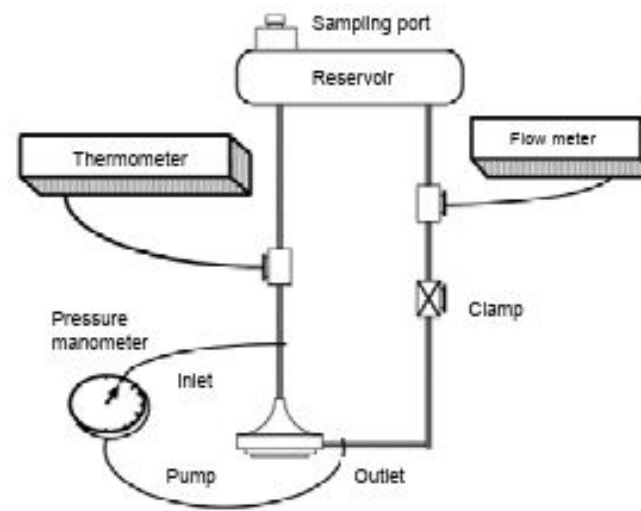


Figure 1. The mock loop circuit for measuring in vivo hemolysis characteristics

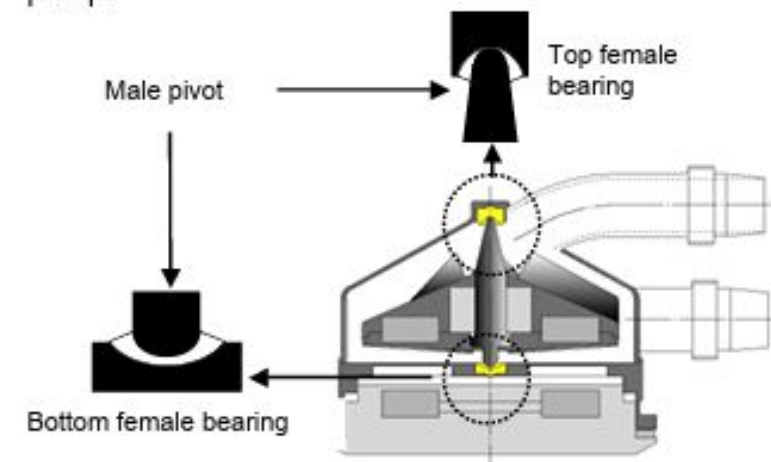


Figure 2. The GYRO blood pump in horizontal top-contact configuration

Operating Condition	RPM	Pressure Head mm of Hg	Flow Rate L/min
A	1600	56.5	3.8
B	2000	100.0	5.0
C	2350	108.0	6.8

Table 1. Three typical operating conditions for experimental and numerical study

Numerical Analysis

Giersiepen *et al.* (Giersiepen 1990) proposed a correlation for % hemolysis in steady simple shear experiments (Figure 3):

$$\frac{\Delta Hb}{Hb} (\%) = 3.62 \times 10^{-5} \sigma^{2.416} \Delta t^{0.785}; \quad \sigma = \mu \frac{du_x}{dy}$$

The hemolysis correlation is applied to the time-dependent complex flows in two ways--"stress-based" and "strain-based". In the stress-based σ is obtained from instantaneous stress-tensor. And in strain-based σ is obtained from instantaneous morphology of Red Blood Cells (Arora 2004). Figure 4 shows the application of stress- and strain-based models in the centrifugal blood pump.

The flow through the pump is simulated with the Deformable-Spatial-Domain/Stabilized-Space-Time (DSD/SST) finite element formulation along with the Shear Slip Mesh-Update Method (SSMUM) (Behr 2002). Figure 5 (a) and (b) show the finite element mesh, and pressure contours, respectively.

One complete revolution of the impeller is simulated and the data is saved at every 18° of impeller rotation. 271 pathlines are traced through the pump. And, stress- and strain-based hemolysis are computed along these pathlines. The one revolutions data is recycled for pathline tracing as shown in Figure 6.

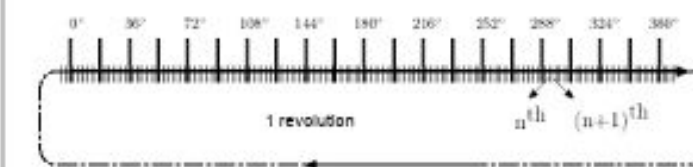


Figure 6. The CFD data and tracer time steps for one revolution

Figure 7 shows the initial distribution of 271 particles. The cumulative hemolysis at the exit is weighted with the flow rate at the initial radial positions.

For operating conditions "A", the 271 particles are released at different impeller angles, and temporal average of the hemolysis is computed.

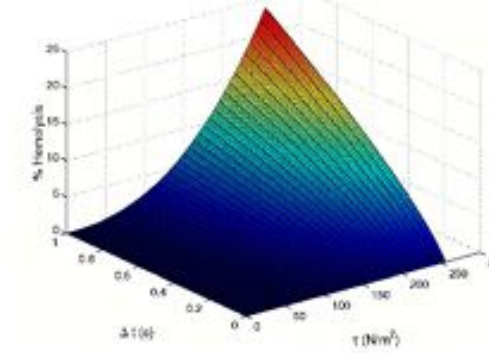


Figure 3. Steady shear stress vs. % hemolysis

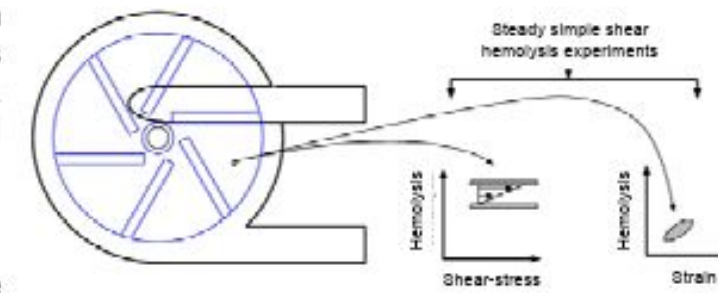
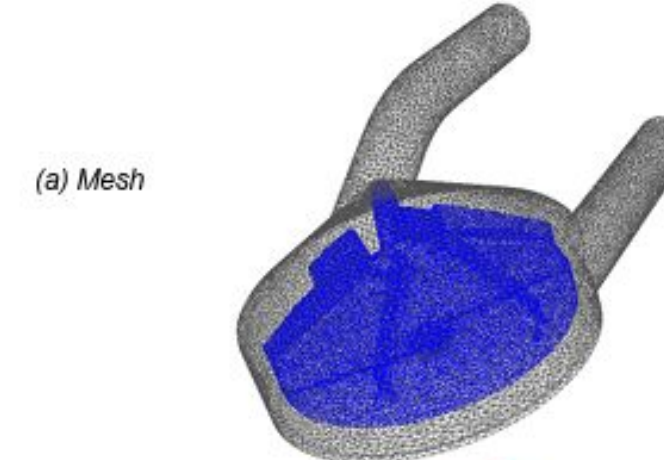
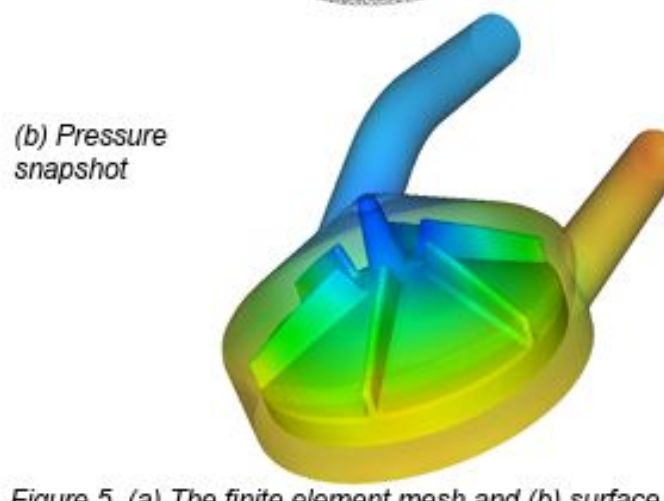


Figure 4. Stress- and strain-based hemolysis predictions for the GYRO blood pump



(a) Mesh



(b) Pressure snapshot

Figure 5. (a) The finite element mesh and (b) surface pressure in the GYRO blood pump; the mesh has 1,017,731 tetrahedral elements and 350,162 space-time nodes

Figure 8 shows a typical pathline through the pump.

Table 2 lists the stress-based and the strain-based predictions of hemolysis in the GYRO blood pump at the three operating conditions.

Operating Condition	Stress-based (NIH)	Strain-based (NIH)	Expt. (NIH)
A	0.378	0.0076	0.0072
B	0.0518	0.0084	0.0070
C	0.0945	0.0139	0.0210

Table 1. Comparison of the stress-based and strain-based predictions of hemolysis with experimentally measured values

Conclusions

The strain-based hemolysis model predictions are very close to the experimentally measured values.

The stress-based method overpredicts hemolysis in the blood pump.

The computational time required for the stress-based and the strain-based hemolysis predictions are comparable.

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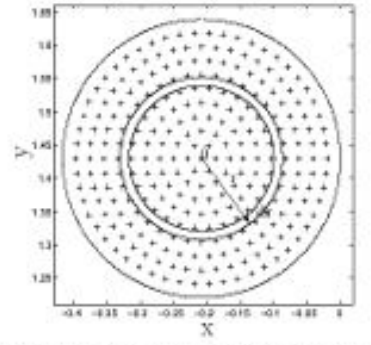


Figure 7. Initial distribution of 271 particles at the pump inlet

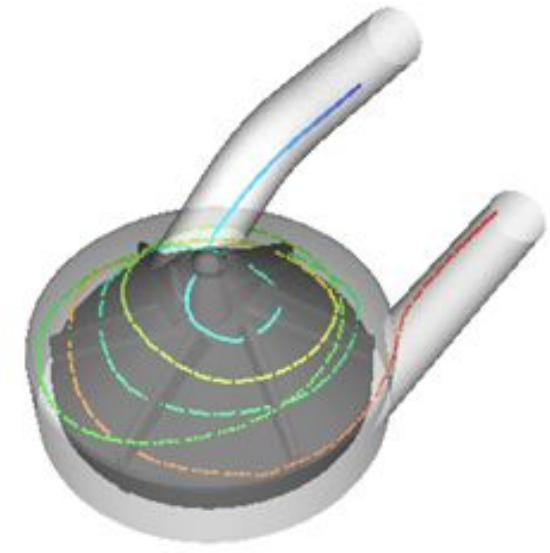


Figure 8. A typical pathline through the blood pump